

dustrial applications? Have a look at some of them:

Test labs for electronics engineers.

A web-enabled RPi and a programmable test and measurement device called the Red Pitaya have come together to create this awesome utility. The Red Pitaya is a portable, programmable, multifunctional test and measurement device. It serves as an oscilloscope, a function generator and a frequency and spectrum analyser, all in one go! The Red Pitaya is connected to a network as a web server, and users can connect to it by simply entering its IP address in the web browser. Interfacing with the RPi allows access to the Red Pitaya console. This application was started at KickStarter and is now shared in a step-by-step approach on DesignSpark.com community by one of their community members.

Command and control microbrewery. The microbrewery business just got a whole lot easier and productive with the RPi-powered brew environment control. It does away with the fancy and expensive equipment that you traditionally need, and instead allows you to brew your first batch of beer with a couple of sensors and a web-enabled RPi. This little set-up allows the brewer to connect to his brewing room from anywhere in the world through the Internet and monitor the temperature and humidity, both of which are very essential to ensure that the beer is drinkable.

Pi joins the Army. There seem to be an increasing number of RPi's featuring in military applications. Reddy explains, "The developer has taken proper care in the PCB design and could not observe any emissions from the traces too. This suggests that the devices and components used on this board are of industrial grade, can sustain extreme conditions and thus can be used for defense applications. The catch here is that boards other than the Raspberry Pi do not pass this test, and so engineers can start using the Raspberry Pi in their applications confidently."

One person from LiveATC has

built an RPi-powered military mode-S logger that automatically decodes mode-S messages through an RTL-SDR dongle using a RPi. Since there are hundreds of messages every second, the RPi takes the brunt of the computing power needed to sift through those messages and log only the new ones to the database. A lot more serious industrial applications can be expected soon with the launch of the modular RPi Compute Module, which specifically targets industrial applications.

RPI projects breaking R&D ground

The RPi's biggest achievement is connected to its roots. Being a board primarily designed for kids to learn with, it inherently becomes the number one choice for interdisciplinary engineers who have very little idea about electronics. What does this mean? It means, scientists and researchers opt for the RPi whenever they need to tinker with electronics.

Going Pi high. The AirPi is an automatic air-quality and weather-monitoring device capable of monitoring information about temperature, humidity, air pressure, light levels, UV levels, carbon monoxide, nitrogen dioxide and smoke level through the Internet. Chasing high-altitude balloons is a sport for those sending RPi GPS-enabled gear to the outer limits of the Earth's atmosphere. Dave Akerman outfitted his balloon's payload with a custom RPi computer that sported a GPS radio, a webcam and various sensors and trackers. His latest set of flights was done at St Alban's School in Ireland on the 7th of June 2014. It flew high with trackers ALBANFLOAT with live image downloads from the sky, and ALBANDATA with a number of environmental trackers.

Life of Pi. A floating research station that is based on top of an autonomous vehicle for long-term data-logging operations, this RPi-powered



BrewPi brewing beer

innovation is definitely a catch. Named the FishPi, it communicates through satellite and is able to make navigational and environmental measurements and observations. Sitting within an unmanned marine surface vessel is the RPi built to cross the Atlantic Ocean while taking scientific measurements. Additionally, a drone project uses the RPi as its sole navigational computer to ensure that the mission is successful.

Seeking crabs. David Soriano, an associate professor of chemistry at the University of Pittsburgh-Bradford, has been using an RPi-controlled webcam to monitor fiddler crabs. As part of the research project, these are being offered thermal polypeptides, rich in the amino-acid tyrosine. Tyrosine starts the pathway to melanin pigment production in the crab, and David is watching for colour changes that result from it.

Pi for social sciences. In the Aix-Marseille Université in France, Sebastien Mathôt, from the Laboratoire de Psychologie Cognitive, has been running his graphical experiment builder, OpenSesame, on RPi. OpenSesame is a graphical, open-source experiment builder for the social sciences, which allows you to build complex experiments with minimum effort. The plug-in framework and Python scripting allow you to incorporate external devices, such as eye trackers, response boxes and parallel port devices, into your experiment. ●

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